Lecture 24.

So far: Expt, S, $E \subseteq S$, $p: S \to [0,1]$, $\mathbb{P}(E)$, $\mathbb{P}(E|F)$, independence

Bayes'

I Independence — II Random Variables — III Linearity of $\mathbb{E}$

↳ $\mathbb{P}(E \cap F) = \mathbb{P}(E)\mathbb{P}(F)$.

Monty Hall Problem

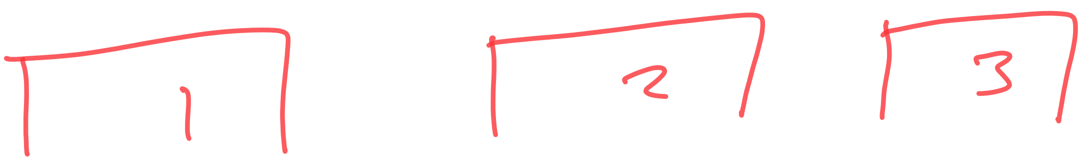
1) A car is placed u.a.r behind one door. Goats are placed behind other doors.

2) You choose a door u.a.r.

3) Monty opens one of the other doors with a goat u.a.r.

expt.

(Q) Should you switch?



$W = \{$You choose the door with the car$\}$.

$M_1 = \{$Monty opens door 1$\}$.

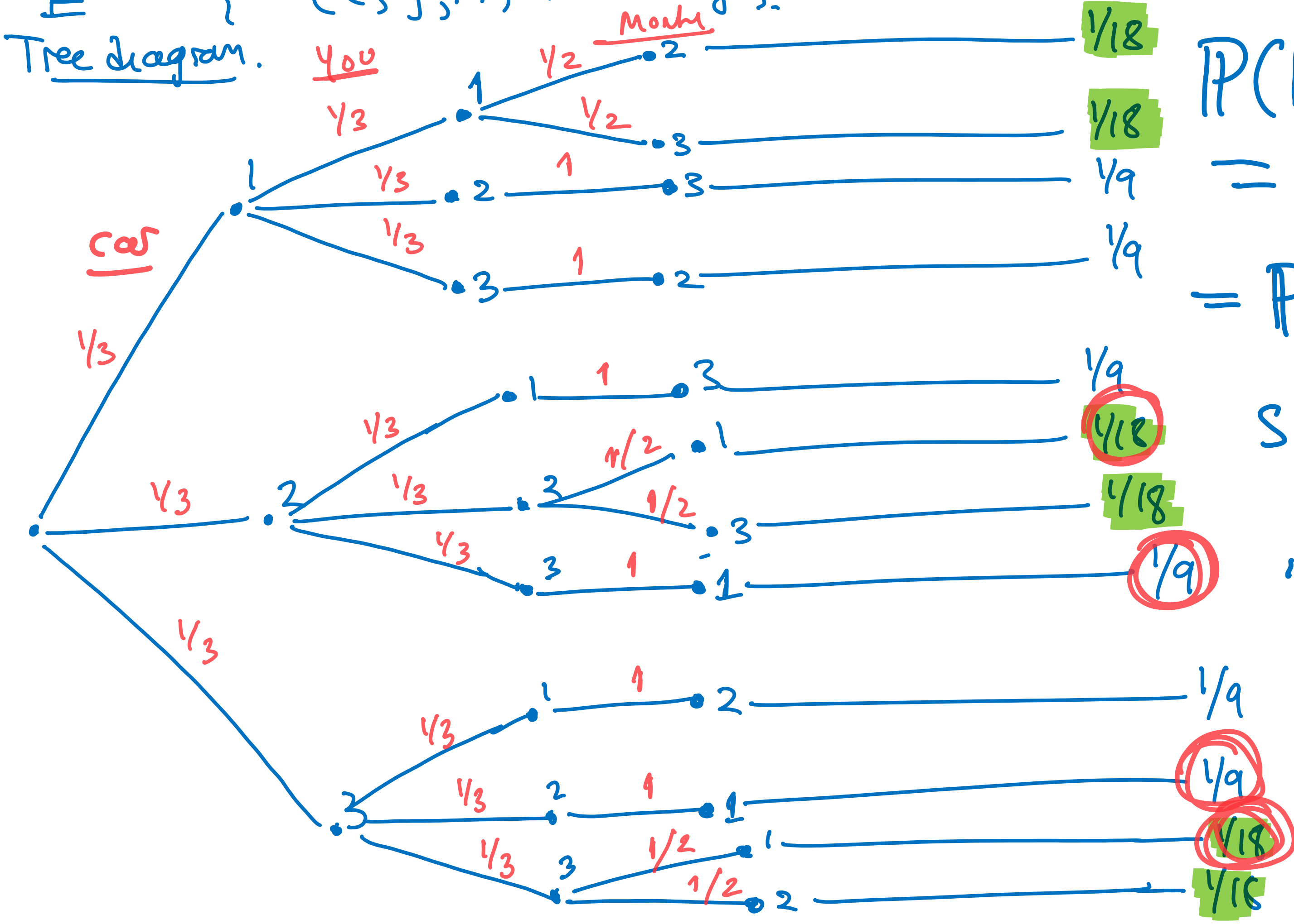
Q) Are W and M_1 independent?

$\mathbb{P}(M_1) = 1/3$.

$\mathbb{P}(W) = 1/3$.

A1) $E = \{(c, y, m) : c = y\}$.

Tree diagram.



$\mathbb{P}(M_1 \cap W) = 1/9 = \mathbb{P}(M_1)\mathbb{P}(W)$

So they are independent.

A2) Define $M_2 = \{\text{Monty chooses \#2}\}$

$M_3 = \{\text{Monty chooses \#3}\}$. (Symmetry)

Observe:

$$\mathbb{P}(M_1|W) = \mathbb{P}(M_2|W) = \mathbb{P}(M_3|W)$$

$$\mathbb{P}(M_1|W) + \mathbb{P}(M_2|W) + \mathbb{P}(M_3|W) = 1.$$

$$\Longrightarrow \boxed{\mathbb{P}(M_1|W)} = \mathbb{P}(M_2|W) = \mathbb{P}(M_3|W) = 1/3.$$

"no door is special"

Also: $\boxed{\mathbb{P}(M_1) = 1/3.}$

$\therefore$ $\mathbb{P}(M_1) = \mathbb{P}(M_1|W)$ So M_1, W are <u>independent</u>

Implies:

$$\mathbb{P}(W|M_1) \overset{\text{Bayes}}{=} \mathbb{P}(M_1|W)\frac{\mathbb{P}(W)}{\mathbb{P}(M_1)} = \frac{1}{3}\cdot\frac{1/3}{1/3} = \underline{\underline{1/3}}$$

$$= \mathbb{P}(W).$$

Very important to define the right events.

(Q) $S = \{$ You win if you switch $\} = \{$ car $\neq$ you, car $\neq$ Monty $\}$.

No door is mentioned here!

Is S independent of M_1?

$$\mathbb{P}(M_1 | S) = \mathbb{P}(M_2 | S) = \mathbb{P}(M_3 | S) = 1.$$

Same argument works.

ex2: Sally Clarke Case

1996: Child #1 died

1997: Child #2 died.

(data)

SIDS: happens to $\frac{1}{8500}$ of the population.

genetic.

Prosecution: $\mathbb{P}(C_1 \text{ dies} \cap C_2 \text{ dies})$

$$= \mathbb{P}(C_1 \text{ dies})\,\mathbb{P}(C_2 \text{ dies})$$

$$= \frac{1}{(8500)^2} = \frac{1}{73 \text{ million}}.$$

Verdict: Guilty.

2003: Royal Stat Society:

C_1, C_2 are not independent.

$$\mathbb{P}(C_2 \mid C_1) > \mathbb{P}(C_2).$$

Mistrial, Released.

II Random Variables

So far: Events, happen or don't.

Now: Assign numerical values to outcomes.

Defn: A random variable on a probability space (S, p) is a function

$$\boxed{X : S \longrightarrow \mathbb{R}.}$$

Not Random
Not a variable

ex: Flip 3 independent coins each with $\underbrace{\mathbb{P}(H) = q}_{\text{biased}}$
$\mathbb{P}(T) = 1-q.$

$S = \{H, T\}^3.$

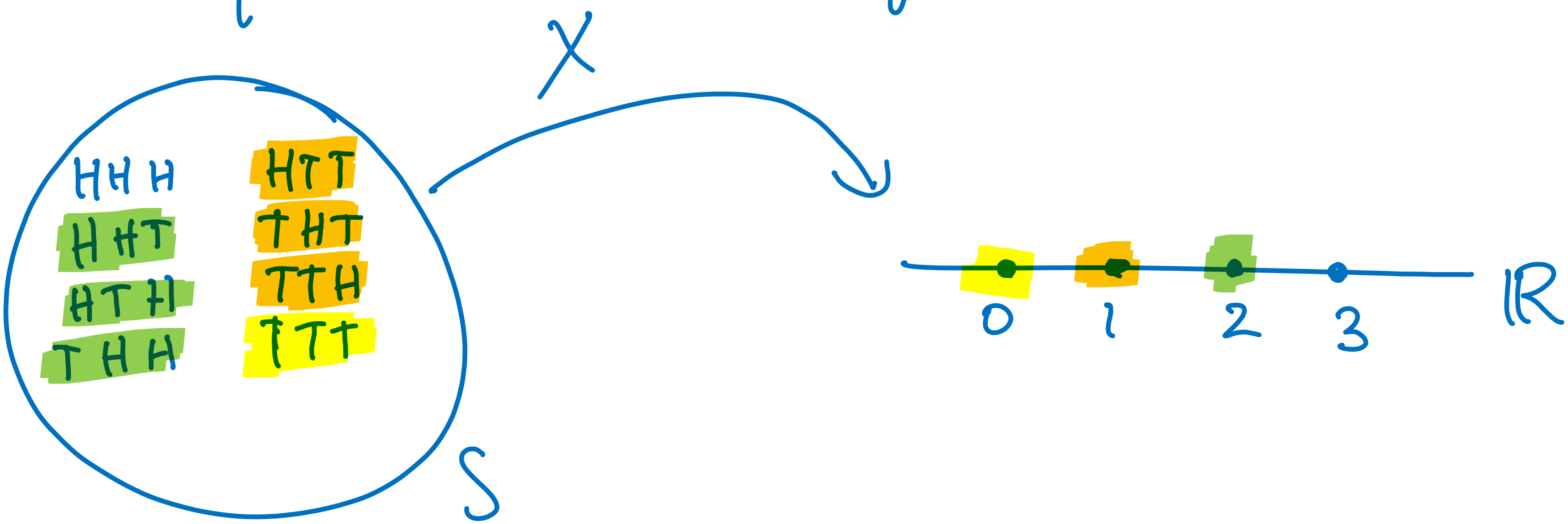
$$p(\underbrace{s_1,s_2,s_3}_{\underline{s}}) \overset{\text{independent}}{=} \mathbb{P}(\text{Coin}_1=s_1)\,\mathbb{P}(\text{Coin}_2=s_2)\,\mathbb{P}(C_3=s_3)$$

$$= q^{\#\text{heads in } \underline{s}}\,(1-q)^{\#\text{tails in } \underline{s}}$$

Define random variables

$X : S \longrightarrow \mathbb{R}$ by $X(s) =$ # of H in s

$Y : S \longrightarrow \mathbb{R}$ by $Y(s) =$ # of T in S.



- X naturally defines the events :

$$\{s : X(s) = 0\} = \{X = 0\}$$

$$\{X = 1\}$$

$$\{X = 2\}$$

$$\{X = 3\}$$

$$\{X = 5\} = \emptyset$$

- X <u>partitions</u> S into

$$X^{-1}(\{0\}),\ X^{-1}(\{1\}),\ X^{-1}(\{2\}),\ X^{-1}(\{3\})$$

where $X^{-1}(A) = \{s \in S : X(s) \in A\}$.

<u>In general</u>: if $X : S \to \mathbb{R}$ is a random variable, we can define events $\{X = r\}$, $r \in \mathbb{R}$ and $S = \bigcup_{r \in \text{range}(X)} \{X = r\}$.

<u>Defn</u>: The <u>expectation</u> of a random variable $X : S \to \mathbb{R}$ is defined as

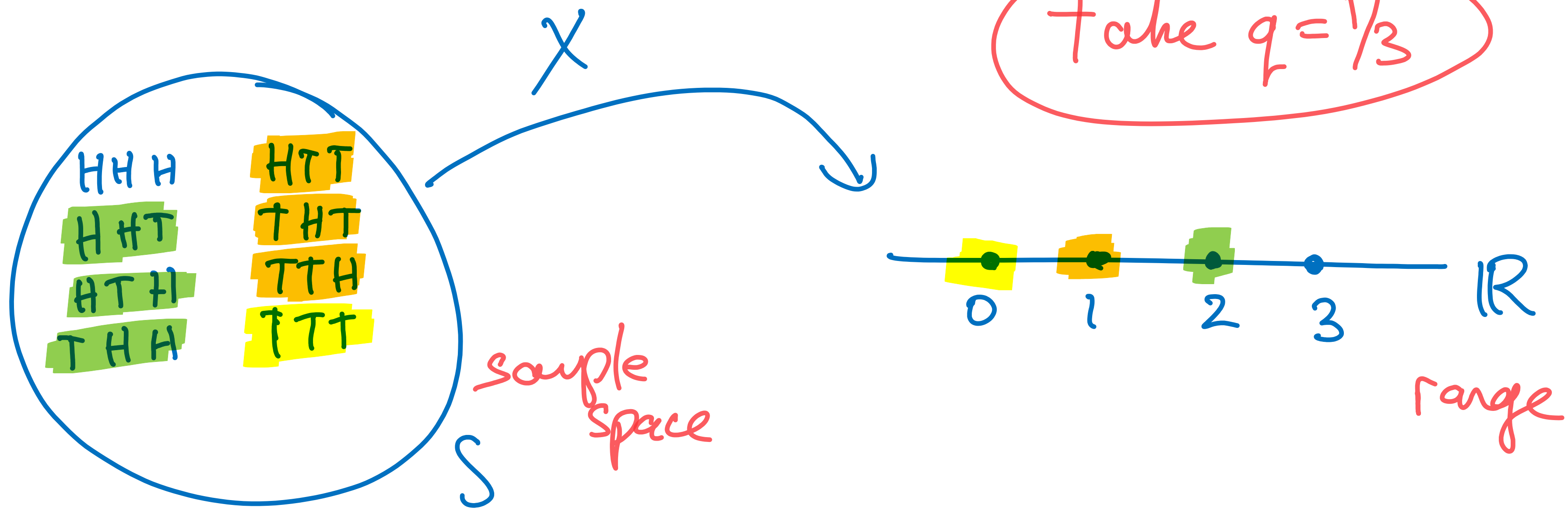
$$\boxed{\mathbb{E}X = \sum_{s \in S} p(s) X(s) = \sum_{r \in \text{range}(X)} \mathbb{P}(X = r) \cdot r}$$

"weighted average over outcomes."

Sum over range(X) weighted by probability.

__ex:__ 3 biased coins w/ bias q $\quad p(s) = q^{\#H(s)}(1-q)^{\#T(s)}$

take $q = 1/3$



__Method 1:__

$$
\begin{aligned}
\mathbb{E}X = {} & p(HHH)\cdot 3 + p(HHT)\cdot 2 + p(HTH)\cdot 2 \\
& + p(THH)\cdot 2 + p(HTT)\cdot 1 + p(THT)\cdot 1 \\
& + p(TTH)\cdot 1 + p(TTT)\cdot 0 \\
= {} & \frac{1}{27}\cdot 3 + \frac{1}{9}\cdot\frac{2}{3}\cdot 2 + \frac{1}{9}\cdot\frac{2}{3}\cdot 2 + \frac{1}{9}\cdot\frac{2}{3}\cdot 2 \\
& + \frac{1}{3}\cdot\left(\frac{2}{3}\right)^2\cdot 1 + \frac{1}{3}\cdot\left(\frac{2}{3}\right)^2\cdot 1 + \frac{1}{3}\left(\frac{2}{3}\right)^2\cdot 1 + \left(\frac{2}{3}\right)^3\cdot 0
\end{aligned}
$$

$$= 1.$$

<u>Method 2</u>: $$\mathbb{E}X = \cancel{0 \cdot \mathbb{P}(X=0)}^{\,0} + 1 \cdot \mathbb{P}(X=1) + 2 \cdot \mathbb{P}(X=2) + 3 \cdot \mathbb{P}(X=3)$$

$$= 1 \cdot 3 \cdot \frac{1}{3} \cdot \left(\frac{2}{3}\right)^2 + 2 \cdot 3 \cdot \frac{1}{9} \cdot \frac{2}{3} + 3 \cdot \frac{1}{27}$$

In general, the <u>Binomial Distribution</u> with n coins and bias $q \in [0,1]$ is defined by:

<u>experiment</u>: flip n independent coins with bias q.

$$p(s) = q^{\#H(s)} (1-q)^{\#T(s)}.$$

$$X : S \longrightarrow \mathbb{R} \text{ by } X(s) = \# \text{ of } H \text{ in } s.$$

then $\boxed{\mathbb{P}(X=r) = \binom{n}{r} q^r (1-q)^{n-r}}$

Reason: There are $\binom{n}{r}$ outcomes with exactly r heads, each with probability $q^r (1-q)^{n-r}$.

Remark: $$\sum_{r=0}^{n} \mathbb{P}(X=r) = \sum_{r=0}^{n} \binom{n}{r} q^r (1-q)^{n-r} \underset{\text{Thm.}}{\overset{\text{Binomial}}{=}} (q+(1-q))^n = 1.$$

Q) What is $\mathbb{E}X$? Method 2: $\sum_{r=0}^{n} r \cdot \binom{n}{r} q^r (1-q)^{n-r}$.

III Linearity of Expectation

We can <u>add</u> two random variables $X, Y : S \to \mathbb{R}$

$$X+Y : S \to \mathbb{R} \text{ by } (X+Y)(s) = X(s)+Y(s)$$

We can <u>scalar multiply</u> $X : S \to \mathbb{R}$ and $a \in \mathbb{R}$

$$aX : S \to \mathbb{R} \text{ by } (aX)(s) = aX(s).$$

< Math 54: the set of random variables on S is a vector space >.

<u>Theorem</u>: If X, Y are random variables on S then

$$\boxed{\mathbb{E}(X+Y) = \mathbb{E}X + \mathbb{E}Y}$$

<u>Corollary</u>:

$$\mathbb{E} \sum_{i \leq n} X_i = \sum_{i \leq n} \mathbb{E}X_i .$$

Proof:

$$\mathbb{E}(X+Y) = \sum_{s\in S} p(s)(X+Y)(s)$$

$$\overset{\text{defn}}{=} \sum_{s\in S} p(s)\left(X(s)+Y(s)\right)$$

$$= \sum_{s\in S} p(s)X(s) + p(s)Y(s)$$

$$\overset{\text{distribute}}{=} \sum_{s\in S} p(s)X(s) + \sum_{s\in S} p(s)Y(s)$$

$$= \mathbb{E}X + \mathbb{E}Y. \qquad \square$$

"The Average of the sum is the sum of the averages."

Recall: n coin flips with bias q. $S=\{H,T\}^n$.

$$X: S \longrightarrow \mathbb{R} \qquad X(s) = \#\text{ of } H.$$

idea: Define n random variables

$$X_i = \begin{cases} 1 & \text{if coin } i \text{ is } H \\ 0 & \text{if coin } i \text{ is } T. \end{cases} \qquad i=1,\dots,n.$$

Zoom in on one coin.

Upshot: $\forall i$: $\mathbb{E}X_i = 0\cdot \cancel{\mathbb{P}(X_i=0)} + 1\cdot \mathbb{P}(X_i=1) = q.$

Observe: $X = X_1 + X_2 + \dots + X_n.$

n coins, hard

By Linearity: $\mathbb{E}X$

1 coin, easy

$= \mathbb{E}X_1 + \mathbb{E}X_2 + \dots + \mathbb{E}X_n$

$= q + q + \dots + q$

$= \underline{\underline{nq}}.$

Key Idea: Decomposed a complicated random variable (X) into a sum of simple ones $(X_1 — X_n)$.